6. 📍 Distance Calculation (CORE LOGIC)

System calculates distance between:

- Victim location `(lat1, lng1)`
- Responder location `(lat2, lng2)`

Using **Haversine Formula**:

$$d = 2r \cdot \arcsin\left(\sqrt{\sin^2\left(\frac{\phi_2-\phi_1}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\lambda_2-\lambda_1}{2}\right)}\right)$$

Where:

- ϕ = latitude
- λ = longitude
- r = Earth radius (~6371 km)

👉 Output:

- Distance for each responder

🚨 AidLink Final System Flow (with Distance Calculation)

1. Victim sends SOS

- User submits:
  - description
  - urgency
  - location `(lat, lng)`

---

2. Backend validates + stores request

- Check location format
- Save into Firestore as:
  - `status = requested`

---

3. AI Emergency Analysis

- AI interprets the situation
- Confirms or refines urgency

---

4. Map urgency → responder role

- **high → emergency**
- **medium → ngo**
- **low → volunteer**

---

5. 🔍 Find eligible responders

Filter responders:

- ✅ verified = true
- ✅ availability = true
- ✅ role matches urgency
- ✅ has valid location

---

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Where:

- ϕ = latitude
- λ = longitude
- r = Earth radius (~6371 km)

👉 Output:

- Distance for each responder

---

7. 🎯 Select nearest responder

- Sort responders by distance
- Pick the **closest one**

Store:

- `assignedResponderId`
- `assignedDistance`

---

8. ❗ If no responder available

- Set request:
  - `status = waiting`
- Keep in queue

---

9. 🚑 Assign responder

- Update:
  - request → `assigned`
  - responder → `availability = false`
  - `activeCases += 1`

---

10. 🔄 Real-time status updates

Responder updates:

- `en_route`
- `arrived`
- `completed`

---

11. 📱 Victim live tracking

- Victim sees:
  - responder assigned
  - movement / ETA (optional)
  - status updates

---

12. ✅ Completion logic

- When done:
  - request → `completed`
  - responder → `activeCases - 1`
  - if 0 → `availability = true`

---

13. 🔁 Auto-assign from queue

- If waiting requests exist:
  - system re-runs dispatch
  - assigns next closest responder

🔥 One-line powerful flow (for presentation)

SOS → AI analysis → urgency mapping → filter responders → calculate distances → assign nearest → real-time tracking → completion → auto-reassign queue

---

💡 Why this is VERY strong for judging

This shows:

- ✅ real algorithm (not just CRUD)
- ✅ intelligent matching (distance-based)
- ✅ scalable dispatch logic
- ✅ fairness (closest responder first)
- ✅ real-world practicality (like Grab/Uber logic)

✅ ✨ ADD THIS SECTION (after "System Flow" or before "Tech Stack")

📍 Intelligent Dispatch & Distance-Based Assignment

AidLink AI does not assign responders randomly. It uses **algorithm-driven dispatch logic** to ensure the fastest and most optimal rescue.

🔍 How it works

When a request is created:

1. The system filters responders who are:
    - ✅ verified
    - ✅ available
    - ✅ matched to urgency level (emergency / NGO / volunteer)
    - ✅ have valid GPS location
2. The system calculates the **real-world distance** between the victim and each eligible responder

3. Responders are ranked by proximity
4. The **nearest responder is automatically assigned**

📐 Distance Calculation

AidLink uses the **Haversine formula** to compute accurate geographic distance on Earth's surface.

This ensures:

- realistic distance (not straight-line map distance)
- better responder selection in real-world terrain

🎯 Why this matters

This approach ensures:

- ⚡ Faster response time
- 📍 Nearest responder prioritization
- ⚖️ Fair and efficient workload distribution
- 🚑 Real-world dispatch behavior (similar to Grab/Uber emergency routing)

🔄 Smart Fallback (Queue System)

If no responders are available:

- The request is **not rejected**
- It is placed into a **waiting queue**
- When a responder becomes available, the system:
  - re-runs the dispatch logic
  - assigns the nearest responder automatically

🚀 OPTIONAL (VERY HIGH IMPACT ADD-ON)

You can also slightly upgrade your **System Flow section** like this:

👉 Replace this line:

> Agent selects most suitable verified responder

👉 With this:

> Agent selects most suitable verified responder using **distance-based ranking (Haversine calculation) and availability filtering**

💡 WHY THIS UPGRADE IS IMPORTANT

Your original README already shows:

- AI ✅
- real-time system ✅
- full workflow ✅

Now with this addition, you show:

- **algorithmic intelligence** ✅
- **optimization logic** ✅
- **engineering depth** ✅

👉 This directly improves:

- *Technical Architecture score*
- *Engineering Execution score*